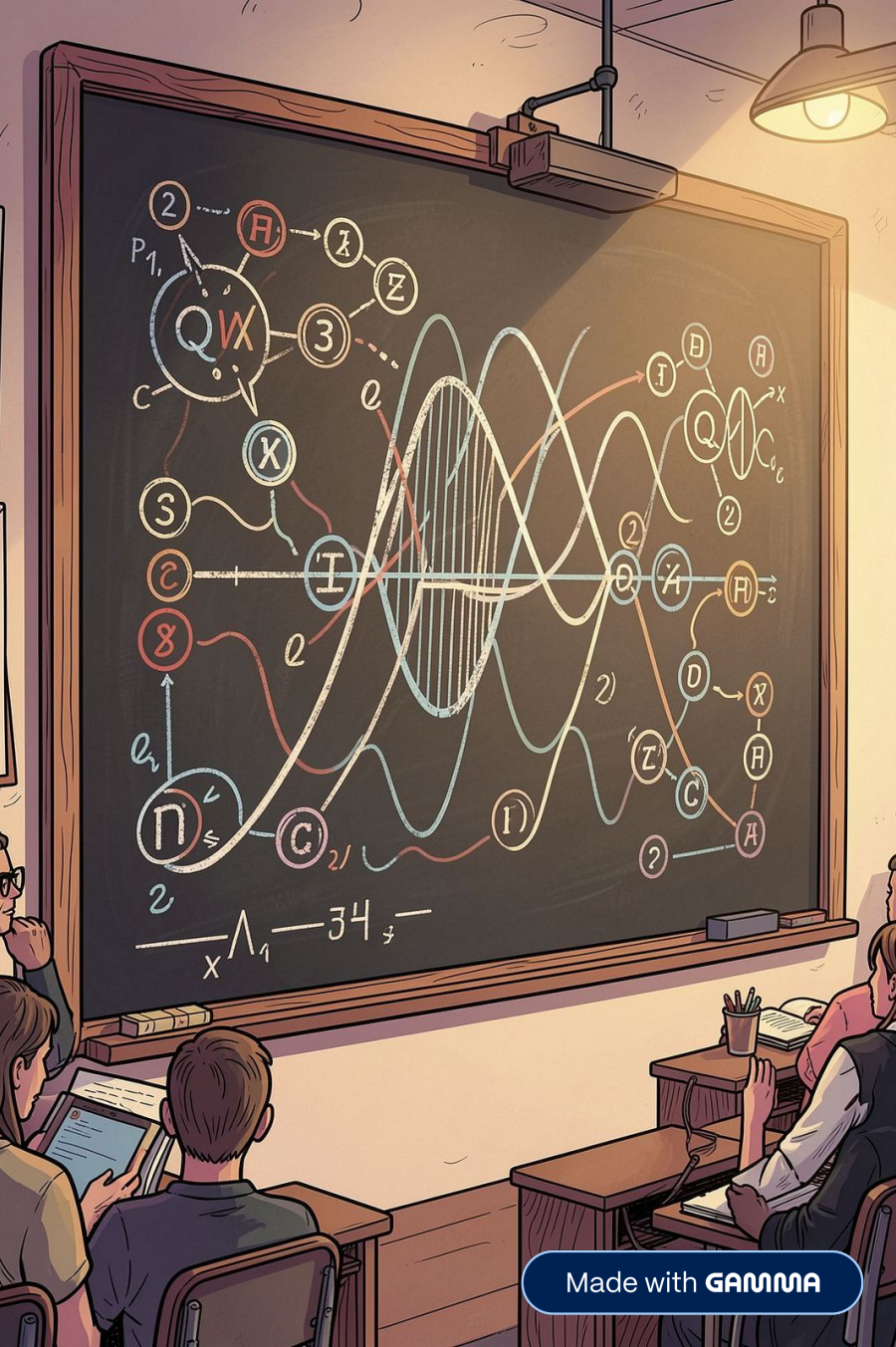
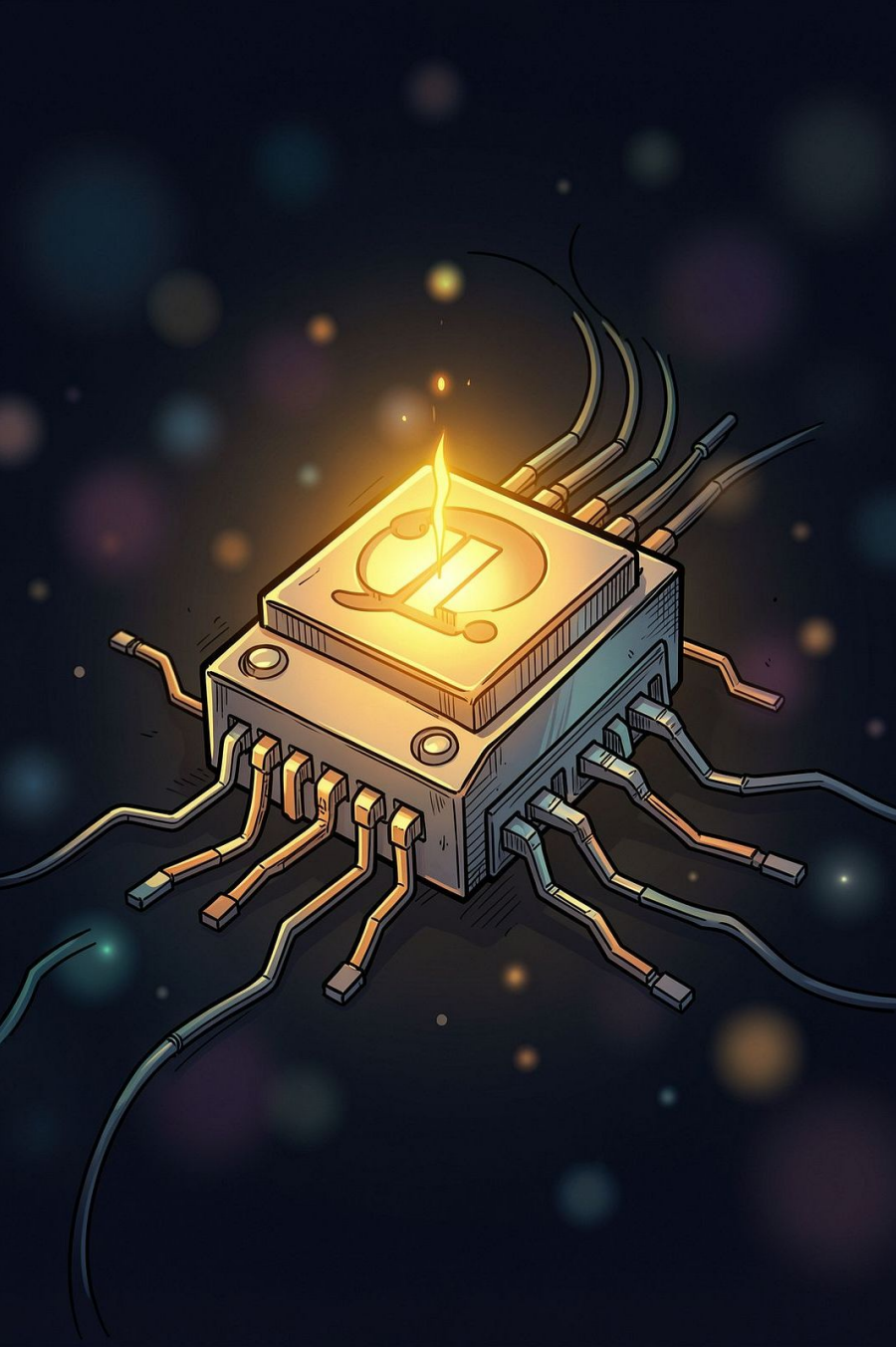


The No-Cloning Theorem





What is the No-Cloning Theorem?

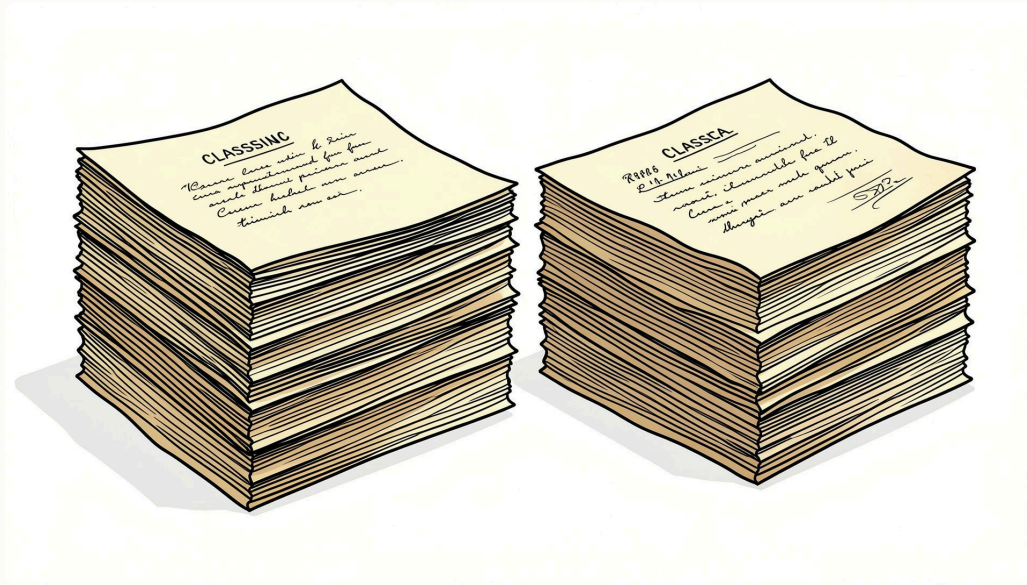
The No-Cloning Theorem is a fundamental result in quantum mechanics: **an arbitrary, unknown quantum state cannot be copied exactly.** Unlike classical bits (which can be duplicated without disturbance), unknown qubits cannot be perfectly duplicated by any universal quantum operation.

Why it matters

Protects quantum information and underlies the security of quantum cryptography.

Consequences

Prevents perfect eavesdropping and enforces fundamental limits on information replication.



Classical vs Quantum Copying

- Classical bits: can be copied perfectly (e.g., 101100 → 101100).
- Quantum qubits: unknown states $|\psi\rangle$ cannot be universally cloned without disturbance.
- Copying a classical string leaves the original unchanged; cloning an unknown qubit is impossible.

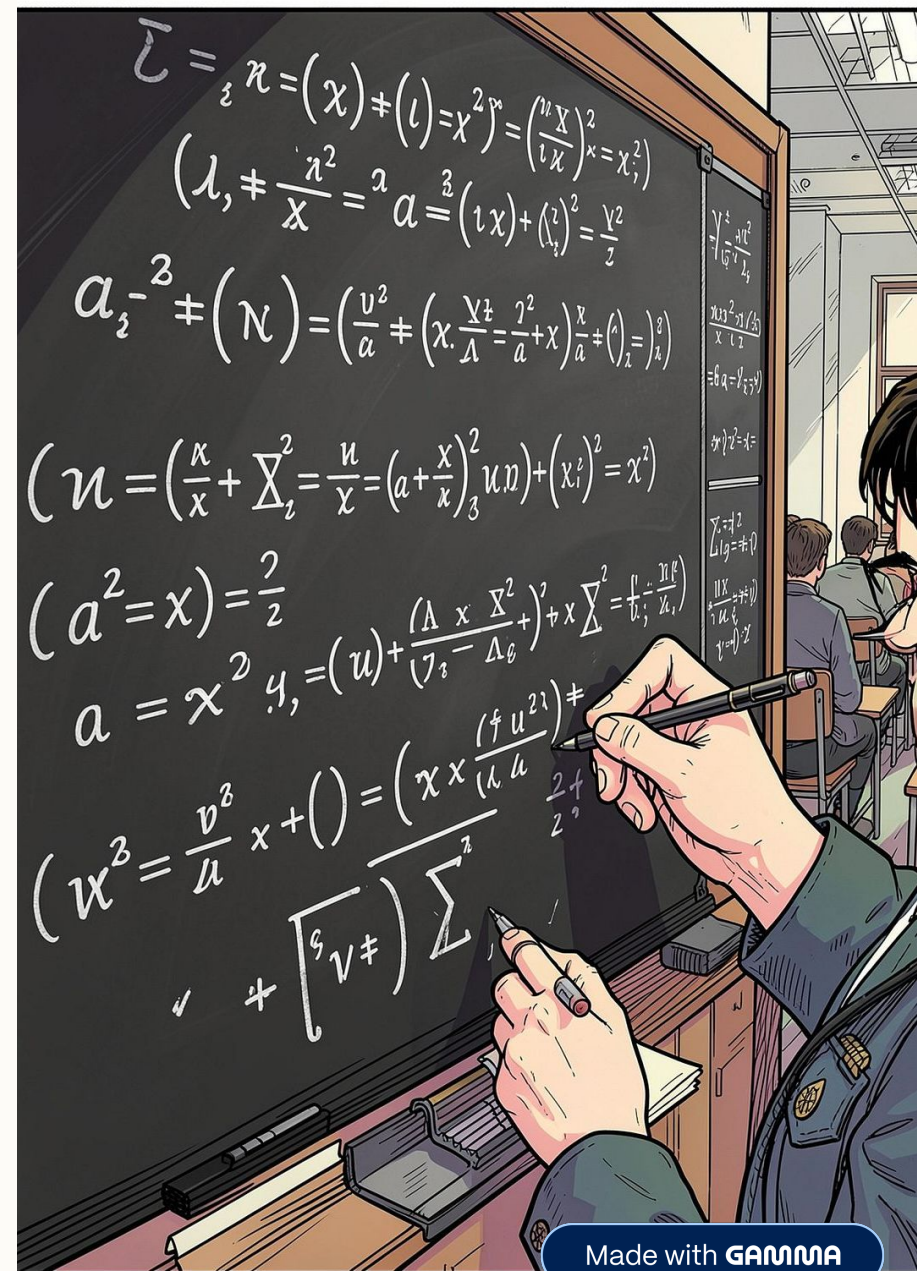
Example: Classical bit copy is trivial; creating $|\psi\rangle|\psi\rangle$ from an arbitrary unknown $|\psi\rangle$ is forbidden.

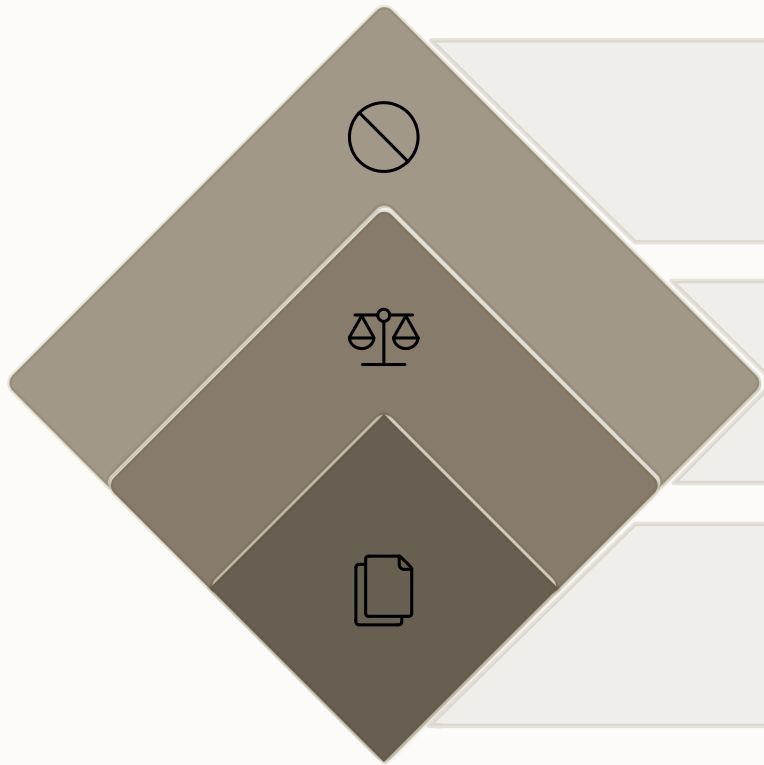
Formal Statement

Suppose a hypothetical universal cloner implements:

$|\psi\rangle|0\rangle \rightarrow |\psi\rangle|\psi\rangle$ for every unknown $|\psi\rangle$. The No-Cloning Theorem says: **no unitary operation U can perform this transformation for all possible $|\psi\rangle$.**

- This is a statement about arbitrary unknown states — not about preparing known states from classical descriptions.





Conclude Contradiction

Only $|\langle \psi | \varphi \rangle| = 0$ or 1

Inner Product Preserved

Unitarity gives $\langle \psi | \varphi \rangle = \langle \psi | \varphi \rangle^2$

Assume Universal Cloner

U clones $|\psi\rangle$ and $|\varphi\rangle$

Sketch: Linearity and unitarity imply inner products must be preserved. If U cloned both $|\psi\rangle$ and $|\varphi\rangle$, then $\langle \psi | \varphi \rangle = \langle \psi | \varphi \rangle^2$, which holds only when $\langle \psi | \varphi \rangle$ is 0 or 1. Thus a universal cloner cannot exist.



Concrete Example

Take $|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. A perfect copier would need to produce $(|0\rangle + |1\rangle)/\sqrt{2} \otimes (|0\rangle + |1\rangle)/\sqrt{2}$ for any input. Because quantum operations are linear, no single unitary can map every input to its duplicated tensor product. Therefore, perfect cloning fails even for simple superpositions.

Practical Implications



Quantum Security

The theorem guarantees that eavesdroppers cannot copy unknown qubits perfectly, forming the foundation of Quantum Key Distribution (QKD).



Secure Communication

Quantum channels can detect interception because attempted measurements or imperfect cloning disturb states.



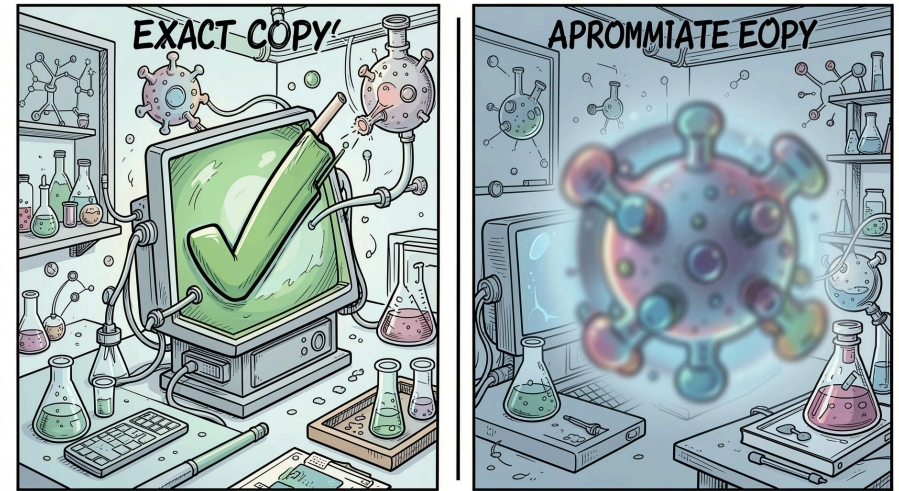
Error Correction & Encoding

Quantum error correction uses encoded redundancy rather than copying unknown qubits directly.



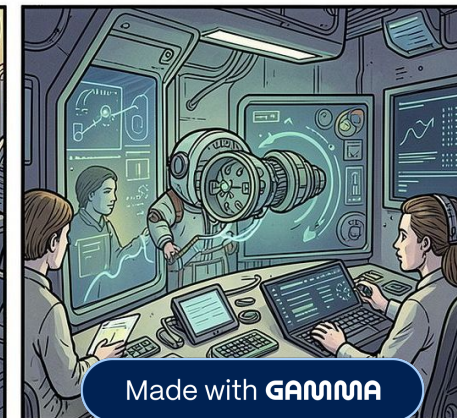
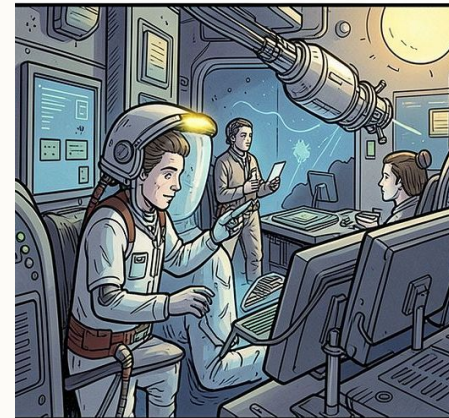
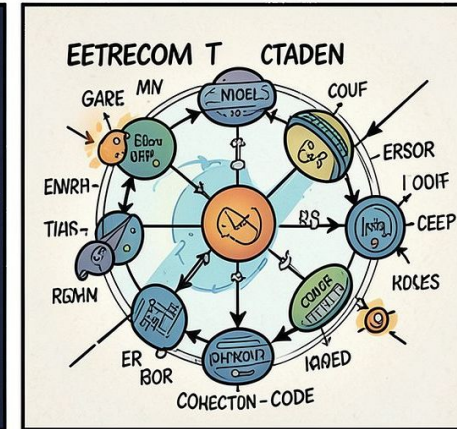
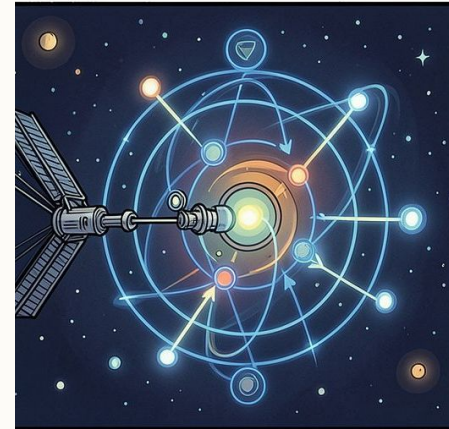
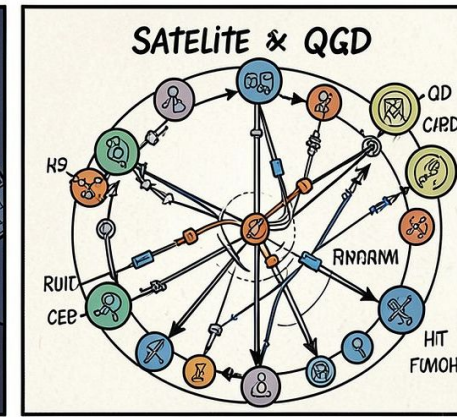
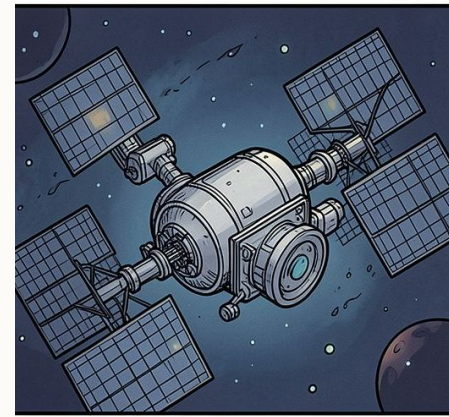
Allowed and Disallowed Actions

- **Disallowed:** Universal perfect cloning of arbitrary unknown states.
- **Allowed:** Preparing multiple copies of a known state (you can re-prepare $|\psi\rangle$ if you know its classical description).
- **Allowed (approximate):** Approximate cloning machines can produce imperfect copies with bounded fidelity — useful in research but never exact.
- **Measurement:** Measuring generally collapses the state and prevents faithful duplication.



Applications & Connections

- Quantum Key Distribution (BB84 and related protocols rely on no-cloning to detect eavesdropping).
- Quantum teleportation transfers an unknown state using entanglement and classical communication — it does not clone.
- Quantum networks and repeaters must use encoding and entanglement swapping rather than copying unknown qubits.



Summary & Takeaways

- The No-Cloning Theorem: unknown quantum states cannot be perfectly copied.
- Derives from linearity and the requirement that quantum evolution is unitary.
- Enables quantum security, shapes communication protocols, and sets fundamental limits for quantum information processing.
- Known states can be re-prepared; approximate cloning exists but is imperfect.

□ Suggested next steps: Work through a formal unitary inner-product proof and explore an approximate cloner fidelity calculation.

